

# Computer Science & IT

## Discrete Mathematics



**Set Theory & Algebra**

**Lecture No. 07**



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# Recap of Previous Lecture



- ✓ **Topic** Number of partitions of a set
- ✓ **Topic** Number of equivalence relation
- ✓ **Topic** Bell number
- ✓ **Topic** Questions



# Topics to be Covered



Topic

GATE PYQ



Topic

Partial order relation



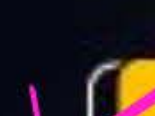
Topic

Partially ordered set (POSET)



Topic

Comparability



Topic

Totally ordered set (TOSET)



Q: Let  $A = \{a, b, c, d, e, f\}$  i.e.  $|A| = 6$

\* How many equivalence relations are possible on set  $A$ , such that cardinality of each equivalence relation is exactly '12'.

$$|A| = 6 \Rightarrow$$

$$\begin{array}{c} 1+1+1+1+1+1 \\ 1+1+1+1+1+1 \end{array} = 7$$

$$\begin{array}{c} 1+1+1+1+2 \\ 1+1+1+1+4 \end{array} = 8$$

$$\boxed{\begin{array}{c} 1+1+1+3 \\ 1+1+1+9 \end{array} = 12}$$

$$\begin{array}{c} 1+1+2+2 \\ 1+1+4+4 \end{array} = 10$$

$$\begin{array}{c} 1+1+4 \\ 1+1+16 \end{array} = 18$$

$$\begin{array}{c} 1+2+3 \\ 1+4+9 \end{array} = 14$$

$$\begin{array}{c} 1+5 \\ 1+25 \end{array} = 26$$

$$\boxed{\begin{array}{c} 2+2+2 \\ 4+4+4 \end{array} = 12}$$

$$\begin{array}{c} 2+4 \\ 4+16 \end{array} = 20$$

$$\begin{array}{c} 3+3 \\ 9+9 \end{array} = 18$$

$$\begin{array}{c} 6 \\ 36 \end{array} = 38$$

No. of  
Equivalence rel<sup>n</sup>

$$\frac{6 \times 5 \times 4}{3!} * {}^3C_3 = 20$$

$$\frac{{}^6C_2 * {}^4C_2 * {}^2C_2}{3!} = 15$$

Total no. of  
required relation

$$= 20 + 15$$

$$= \boxed{35}$$



What is the possible number of reflexive relations on a set of 5 elements?



25 Total

5 diagonal

$(25-5) = \text{Non-diagonal}$

$$= {}^5C_5 * 2^{25-5}$$

$$= 2^{20}$$

A

$2^{10}$

B

$2^{15}$

C

$2^{20}$

D

$2^{25}$

For a set  $A$ , the power set of  $A$  is denoted by  $2^A$ . If  $A = \{5, \{6\}, \{7\}\}$ , which of the following options are TRUE?



☒ (1)  $\emptyset \in 2^A$

☒ (2)  $\emptyset \subseteq 2^A$

☒ (3)  $\{5, \{6\}\} \in 2^A$

☒ (4)  $\{5, \{6\}\} \subseteq 2^A$

☐ A (1) and (3) only

☐ B (2) and (3) only

☒ C (1), (2) and (3) only

☐ D (1), (2) and (4) only

Let  $R$  be the relation on the set of positive integers such that  $aRb$  if and only if  $a$  and  $b$  are distinct and have a common divisor other than 1. Which one of the following statements about  $R$  is True?

Can not be reflexive  
(it will be irreflexive)

Symmetric ✓

Reflexive ✗

Transitive ✗

- A**  $R$  is symmetric and reflexive but not transitive
- B**  $R$  is reflexive but not symmetric and not transitive
- C**  $R$  is transitive but not reflexive and not symmetric
- D**  $R$  is symmetric but not reflexive and not transitive

let  $(a,b) \in R \wedge (b,a) \in R$   
because Symmetric

then  $(a,a) \in R$

but  $(a,a) \notin R$  because  $a=a$



The cardinality of the power set of  $\{0,1,2,\dots,10\}$  is\_\_\_\_\_.



$$2^{11} = \underline{\underline{2048}}$$



Suppose  $U$  is the Power set of the set  $S = \{1, 2, 3, 4, 5, 6\}$ . For the any  $T \in U$ , let  $|T|$  denote the number of elements in  $T$  and  $T'$  denote the complement of  $T$ . For any  $T, R \in U$ , let  $T \setminus R$  be the set of all elements in  $T$  which are not in  $R$ . Which one of the following is true?

$T$  is a subset of  $S$

will return false

Max  $|X \cap Y| = 5$   
Min  $|X \cap Y| = 4$

Set difference ( $T - R$ )

it can not be '0'

~~A~~

$\forall X \in U, (|X| = |X'|)$

~~B~~

$\exists X \in U, \exists Y \in U, (|X| = 5, |Y| = 5 \text{ and } X \cap Y = \emptyset)$

~~C~~

$\forall X \in U, \forall Y \in U, (|X| = 2, |Y| = 3 \text{ and } X \setminus Y = \emptyset)$

D

$\forall X \in U, \forall Y \in U, (X \setminus Y = Y' \setminus X')$

$X = \{1, 2, 4, 6\}$

$X' = \{3, 5\}$

$Y = \{1, 3, 5\}$

$Y' = \{2, 4, 6\}$

$Y' \setminus X' = \{2, 4, 6\}$

$X - Y = \{2, 4, 6\}$





Let  $R$  be a relation on the set of ordered pairs of positive integers such that  $((p, q), (r, s)) \in R$  if and only if  $p - s = q - r$ . Which one of the following is true about  $R$ ?

$R$  is defined from set  $N \times N$  to  $N \times N$

$$(p, q) R (r, s) \text{ iff } (p - s) = (q - r)$$

$$(a, b) R (a, b) \text{ iff } (a - b) = (b - a)$$

- ☐ A Both reflexive and symmetric
- ☐ B Reflexive but not symmetric
- ☒ C Not reflexive but symmetric
- ☐ D Neither reflexive nor symmetric

It need not be true for all  $a \& b$

(- , -)  
(- , -)  
(- , -)  
(- , -)

$\therefore$  Not reflexive

if  $(p, q) R (r, s)$  then  $(r, s) R (p, q)$   
iff  $(p - s) = (q - r)$  then  $(r - q) = (s - p)$   $\therefore$  Symmetric  
(if it is true  $\Rightarrow$ )



A binary relation  $R$  on  $N \times N$  is defined as follows:  $(a, b) R (c, d)$



if  $a \leq c$  or  $b \leq d$ . Consider the following propositions:

✓ **P:**  $R$  is reflexive

✗ **Q:**  $R$  is transitive

∴ Not transitive

Which one of the following statements is TRUE

- A** Both  $P$  and  $Q$  are true
- B**  $P$  is true and  $Q$  is false
- C**  $P$  is false and  $Q$  is true
- D** Both  $P$  and  $Q$  are false

$(p, q) R (p, q) \in R$  or not?

$p \leq p$  (or)  $q \leq q$   
 $a \leq c$  True  $b \leq d$  True

∴  $(p, q) R (p, q), \forall (p, q) \in N \times N$

∴ Reflexive

then is false

$10 \neq 3$   $50 \neq 20$

$10 \leq 20$   $50 \neq 10$   
 $(a_1 \leq a_2 \text{ or } b_1 \leq b_2)$

$20 \neq 3$   $10 \leq 20$   
 $(a_2 \leq a_3 \text{ or } b_2 \leq b_3)$

then  $(a_1 \leq a_3 \text{ or } b_1 \leq b_3)$

if  $((a_1, b_1) R (a_2, b_2) \wedge (a_2, b_2) R (a_3, b_3))$  then  $(a_1, b_1) R (a_3, b_3)$

$(a_2, b_2) R (a_3, b_3)$

$x R y \wedge y R z$  then  $x R z$



Let  $G$  be an arbitrary group.

Consider the following relations on  $G$ :

$R_1: \forall a, b \in G, aR_1b$  if and only if  $\exists g \in G$  such that  $a = g^{-1}bg$

$R_2: \forall a, b \in G, aR_2b$  if and only if  $a = b^{-1}$

Which of the above is/are equivalence relation/relation(s)?

- A**  $R_1$  and  $R_2$
- B**  $R_1$  only
- C**  $R_2$  only
- D** Neither  $R_1$  nor  $R_2$

After  
group  
theory

Let  $R$  be the set of all binary relations on the set  $\{1,2,3\}$ . Suppose a relation is chosen from  $R$  at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is \_\_\_\_\_.



ie.  $R$  is of size =  $2^{(3 \times 3)} = 2^9 = 512$

$$\text{Probability} = \frac{\text{Favourable Event}}{\text{Total Outcomes}}$$

No. of reflexive  
rel<sup>n</sup> in set  $R = 2^{3^2-3} = 2^6 = 64$

$$= \frac{64}{512} = \frac{2^6}{2^9} = \frac{1}{8}$$

$$= \underline{\underline{0.125}}$$





A relation  $R$  is said to be circular if  $aRb$  and  $bRc$  together imply  $cRa$ . Which of the following options is/are correct?

~~A~~

If a relation  $S$  is reflexive and symmetric, then  $S$  is an equivalence relation.

transitive = ?

~~B~~

If a relation  $S$  is circular and symmetric, then  $S$  is an equivalence relation.

Empty will also be circular & Empty rel<sup>n</sup> is symmetric  $\Rightarrow$  But Empty rel<sup>n</sup> is not reflexive  $\therefore$  Can not be Equivalence Rel<sup>n</sup>

C

If a relation  $S$  is reflexive and circular, then  $S$  is an equivalence relation.

~~D~~

If a relation  $S$  is transitive and circular, then  $S$  is an equivalence relation.

No, Empty relation is transitive & Circular but not reflexive  $\therefore$  Not an Equivalence rel<sup>n</sup>.

Let  $A = \{a, b, c, d\}$

$S = \{(a,a), (b,b), (c,c), (d,d), \underline{(a,b)}, (b,a)\}$

Equivalence

Reflexive ✓

Symmetric ✓

Transitive ✓

$(a,b) \notin (b,b) \Rightarrow$  then because  $(b,a) \in S$   
Circular

Hence if Reflexive & Circular,  
then Symmetric as well

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Let  $\underline{(a,b)} \in S$  &  $\underline{(b,c)} \in S$  then  $(a,c) \in S$

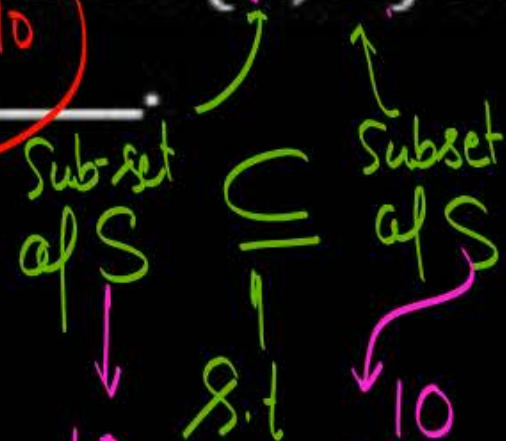
By Circular  
 $\therefore \underline{(c,a)} \in S$  & we know  $\underline{(a,a)} \in S$   
circular  
 $\underline{(a,c)} \in S$

Hence transitive





Let  $S$  be a set consisting of 10 elements. The number of tuples of the form  $(A, B)$  such that  $A$  and  $B$  are subsets of  $S$ , and  $A \subseteq B$  is  $3^{10}$ .



$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

No. of such tuple of type  $(A, B)$  possible

$$= {}^{10}C_0 * 2^{10} + {}^{10}C_1 * 2^9 + {}^{10}C_2 * 2^8 + \dots + {}^{10}C_{10} * 2^0$$

$$(1+x)^n = {}^nC_0 x^0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \dots + {}^nC_n (x)^n$$

$$= {}^nC_0 x^0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \dots + {}^nC_n (x)^n$$

$$(1+2)^{10} = {}^{10}C_0 (2)^0 + {}^{10}C_1 (2)^1 + {}^{10}C_2 (2)^2 + \dots + {}^{10}C_{10} (2)^{10}$$

$$= 3^{10}$$

$$\begin{aligned} &\checkmark |A| \text{ of size } = 0 \Rightarrow {}^{10}C_0 * 2^{10-0} \\ &+ \\ &\checkmark |A| \text{ of size } = 1 \Rightarrow {}^{10}C_1 * 2^{10-1} \\ &+ \\ &\checkmark |A| \text{ of size } = 2 \Rightarrow {}^{10}C_2 * 2^{10-2} \\ &+ \\ &\checkmark |A| \text{ of size } = 3 \Rightarrow {}^{10}C_3 * 2^{10-3} \\ &+ \\ &\vdots \\ &\checkmark |A| \text{ of size } = 10 \Rightarrow {}^{10}C_{10} * 2^{10-10} \end{aligned}$$



Let  $S$  be a set consisting of 10 elements. The number of tuples of the form  $(A, B)$  such that  $A$  and  $B$  are subsets of  $S$ , and  $A \subseteq B$  is \_\_\_\_\_.



$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 3^{10}$$

If we need  $A \subseteq B$  in  $(A, B)$

then for each element of set  $A$  we have '3' choices

- ① Present in  $A$  & Present in  $B$
- ② Not present in  $A$  & Present in  $B$
- ③ Not present in  $A$  & Not present in  $B$

~~Present in  $A$  & Not present in  $B$  is not allowed~~





A pennant is a sequence of numbers, each number being 1 or 2. An  $n$ -pennant is a sequence of numbers with sum equal to  $n$ . For example,  $(1,1,2)$  is a 4-pennant. The set of all possible 1-pennants is  $\{(1)\}$ , the set of all possible 2-pennants is  $\{(2), (1,1)\}$  and the set of all 3-pennants is  $\{(2,1), (1,1,1), (1,2)\}$ . Note that the pennant  $(1,2)$  is not the same as the pennant  $(2,1)$ . The number of 10-pennants is 89.

Partial order relation  
&  
POSET





## Topic : Partial Order Relation

- A relation  $R$  on set  $A$  is called a Partial order relation if and only if relation  $R$  is Reflexive, anti-symmetric and transitive

Partial order = Reflexive + Anti-symmetric + Transitive

eg. Let  $A = \{1, 2, 3\}$

$$\Delta_A = \{(1,1), (2,2), (3,3)\}$$

Reflexive ✓

Anti-symmetric ✓

Transitive ✓

∴ it is a  
Partial order  
relation

✓  $\Delta_A$  is the smallest partial order relation  
on any set  $A$ .



\* Diagonal relation on set  $A$  is the only relation which is an Equivalence relation as well as a Partial order relation

eg. Let  $A = \{1, 2, 3\}$

$$R = \{ \underbrace{(1,1), (2,2), (3,3)}_{\text{Reflexive}}, \underbrace{(1,2), (1,3), (2,3)}_{\text{transitive}} \}$$

Anti symmetric

Reflexive

transitive

∴ Partial order relation





## Topic : Partially Ordered Set

[POSET]



- Let  $R$  is a partial order relation on set  $A$ .
- Set  $A$  along with the partial order relation  $R$  defined on set  $A$  is called a POSET denoted by  $(A, R)$ .

Note: If it is given that  $(A, R)$  is a POSET,  
then ' $A$ ' is a set, and  
' $R$ ' is a partial order relation on set  $A$ .

$A =$  Any set of real no.s.

Relation

$\leq$   
↑

Reflexive  $\Rightarrow a \leq a$  is true for all real no.s  
 $\therefore$  Reflexive

Anti-symmetric  $\Rightarrow$   $\boxed{a \leq b \ \& \ b \leq a \text{ iff } a = b}$   
if  $a \neq b$  then Either  $a \not\leq b$   
or  $b \not\leq a$   
or  $a \not\leq b \ \& \ b \not\leq a$  }  $\therefore$  Anti-symmetric

transitive if  $a \leq b \ \& \ b \leq c$   
then  $a \leq c \therefore$  transitive



Note:

Let 'A' is any set of real numbers,  
then  $(A, \leq)$  is a POSET

A is any set of positive integers.

Relation  $\div$   
divides

$(a, b) \in \div$  iff  
a divides b

① Reflexive: Every +ve integer divides itself  
 $\therefore$  Reflexive

② Anti-symmetric:  $(a \text{ divides } b) \wedge (b \text{ divides } a) \text{ iff } a=b$   
if  $a \neq b$  then either  $a \nmid b$   
or  $b \nmid a$   
or  $a \nmid b \wedge b \nmid a$  }  $\therefore$  Anti-symmetric

③ Transitive: if  $a \div b \wedge b \div c$   
then  $a \div c \therefore$  transitive



Note:

Let 'A' is any set of positive integers <sup>ie,  $0 \notin A$</sup>

then  $(A, \div)$  is a POSET

{where relation ' $\div$ ' denotes "divides" }

don't  
confuse  
with  
"divided by"

Note:

Let  $C$  is any collection of sets,  
then  $(C, \subseteq)$  is a POSET.  
eg.  $C = \{S_1, S_2, \dots\}$  where  $S_1, S_2, \dots$  etc all are sets.  
subset (is subset of)

$$(X, Y) \in \subseteq \\ \text{iff } X \subseteq Y$$

① Reflexive because Every set is a subset of itself.

② Anti-symmetric :  $(S_1 \subseteq S_2 \ \& \ S_2 \subseteq S_1) \text{ iff } S_1 = S_2$   
if  $S_1 \neq S_2$  then either  $S_1 \not\subseteq S_2$   
or  $S_2 \not\subseteq S_1$

③ Transitive : if  $S_1 \subseteq S_2 \ \& \ S_2 \subseteq S_3$  or  $S_1 \not\subseteq S_2 \ \& \ S_2 \not\subseteq S_1$   
then  $S_1 \subseteq S_3 \therefore$  transitive





## Topic : Comparability

• Let  $(A, R)$  be a POSET,

• Let  $a, b \in A$ ,

Elements 'a' and 'b' are said to be Comparable  
if and only if,

$$a^R b \quad \underline{\text{or}} \quad b^R a$$

$$\underline{\text{i.e.}}, (a, b) \in R \quad \underline{\text{or}} \quad (b, a) \in R$$

Let  $a, b \in A$ ,

If neither  $(a, b) \in R$  nor  $(b, a) \in R$ , then  $a$  &  $b$  are not Comparable w.r.t. POSET  $(A, R)$

→ In a POSET  $(A, R)$ ,  
 $R$  is a partial order relation,  
i.e.,  $R$  is guaranteed to be reflexive  
∴  $(a, a) \in R \quad \forall a \in A$

Hence, Every element of set  $A$  is  
always comparable with itself.



eg. Let  $A = \{1, 2, 3, 4\}$   
And we know  $(A, \div)$  is a POSET.

Pair of two elements are

$1 \nless 2$

$(1, 2) \in \div$

$\therefore 1 \nless 2$

are  
Comparable

$1 \nless 3$

$(1, 3) \in \div$

$\therefore 1 \nless 3$

are  
Comparable

$1 \nless 4$

$(1, 4) \in \div$

$\therefore 1 \nless 4$

are  
Comparable

$2 \nless 3$

Neither  $(2, 3) \in \div$

nor  $(3, 2) \in \div$

$\downarrow$

$\therefore 2 \nless 3$  are  
not Comparable  
w.r.t. given POSET

$2 \nless 4$

$(2, 4) \in \div$

$\therefore 2 \nless 4$

are  
Comparable

$3 \nless 4$

Neither  $(3, 4) \in \div$   
nor  $(4, 3) \in \div$

$\downarrow$

$\therefore 3 \nless 4$   
are not  
Comparable

Note:

In a POSET  $(A, R)$ ,

Every pair of elements of set  $A$  need  
not be comparable





## Topic : Totally Ordered Set

/ (TOSSET)



• Let  $(A, R)$  be a POSET,

If Every pair of elements of set  $A$   
are comparable, then POSET  $(A, R)$   
is called a TOSSET, and relation ' $R$ '  
can be called a "Total order relation"

Note :-

- ① Every TOSET is also a POSET,  
But Every POSET need not be a TOSET

Similarly, ② Every Total order relation is also a  
Partial order relation, but Every Partial  
Order relation need not be a Total  
Order relation.



eg: Let  $A_1 = \{1, 2, 3, 4\}$   
and  $(A_1, \div)$  is a POSET



But it is not a TOSET,  
because there exists some  
pair of elements that  
are not comparable

eg: Let  $A_2 = \{1, 2, 4, 8\}$   
and  $(A_2, \div)$  is a POSET.

$(1, 2)$   $(1, 4)$   $(1, 8)$   $(2, 4)$   $(2, 8)$   $(4, 8)$

$(1, 2) \in \div$   $(1, 4) \in \div$   $(1, 8) \in \div$   $(2, 4) \in \div$   $(2, 8) \in \div$   $(4, 8) \in \div$

Every pair of element is comparable

$\therefore$  POSET  $(A_2, \div)$  is also a TOSET.



## 2 mins Summary



✓  
Topic

GATE PYQ

✓  
Topic

Partial order relation

✓  
Topic

Partially ordered set (POSET)

✓  
Topic

Comparability

✓  
Topic

Totally ordered set (TOSET)



**THANK - YOU**